

Ubuntu Screened Subnet Workstation

Cyber Range Build — Phase 1 (Paused)

Date: *[enter date]*

System Role: Screened / Monitoring Workstation

Environment: Home Cyber Range (secure-by-design / vulnerable-by-choice)

Objective

The goal of this phase was to prepare an **Ubuntu client workstation** for use as a **screened subnet monitoring node** on the default VLAN. This machine will eventually operate with **two NICs** and serve as a **visibility and defense platform** for observing traffic, analyzing security events, and supporting blue-team style labs.

This phase focused on **tool installation only**.

Domain integration was intentionally deferred to a later phase.

Network Design Intent

- **NIC 1:** Management / Domain VLAN (Active Directory access)
 - **NIC 2:** Default / Screened VLAN (IoT, user, and test traffic)
 - Purpose:
 - Passive traffic capture
 - Network discovery
 - IDS / monitoring
 - Defensive security demonstrations
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Tools Installed

Core System Preparation

- System updated and upgraded
- Base networking and diagnostic utilities installed
- Package management verified

Network Monitoring & Discovery

- **Wireshark**
 - Installed and configured for non-root packet capture
 - User added to `wireshark` group
- **Netdiscover**
 - Installed for ARP-based discovery on screened VLAN

- **Nmap**
 - Installed for host and service enumeration
- **Tcpdump / Tshark**
 - Installed for CLI packet capture and analysis

Intrusion Detection & Logging

- **Suricata**
 - Installed and enabled as a network IDS
 - Logs available under `/var/log/suricata`
- **Auditd**
 - Installed for host-level auditing

Host Hardening & Defense

- **UFW (Uncomplicated Firewall)**
 - Enabled
 - SSH allowed
 - Inbound traffic restricted by default
 - **Fail2ban**
 - Installed for brute-force mitigation
 - **Lynis**
 - Installed for system security auditing and baseline assessment
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SSH Access Configuration

- OpenSSH Server installed
- SSH service enabled and running
- Firewall rule added to allow SSH access
- (Optional hardening steps identified but deferred)

This allows remote administration of the monitoring workstation from the management network.

Tools Not Installed (Deferred)

- **BloodHound**
 - Installation was paused due to package availability and dependency considerations
 - Neo4j and BloodHound will be addressed in a later session when Active Directory integration begins

This pause was intentional to avoid partial or misconfigured AD tooling before domain join.

Current Status

- ✓ System prepared and updated
- ✓ Monitoring and defense tools installed
- ✓ SSH access enabled and verified
- ✓ Firewall enabled
- ✓ Ready for domain integration

II Paused before:

- Domain join (`realmd`, `sssd`)
 - BloodHound / Neo4j installation
 - AD authentication testing
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Next Steps (Planned)

1. Install and configure **BloodHound + Neo4j**
 2. Join Ubuntu workstation to the Active Directory domain
 3. Validate domain authentication
 4. Bind monitoring tools to the screened NIC only
 5. Add IDS tuning rules for IoT / default VLAN
 6. Document findings and capture screenshots for lab artifacts
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Notes

This workstation is intentionally designed as a **hybrid security node**:

- Defensive by default
- Capable of demonstrating insecure configurations when required
- Used for education, blue-team labs, and cyber-range visibility

All actions in this phase were performed with the understanding that this environment is **not production**, but a controlled learning range.

End of Phase 1 — Paused for continuation after class